Mode A (t = 5.0 ms) Inner-Loop Verification Report 3 (Fresh Bitstream Flash)

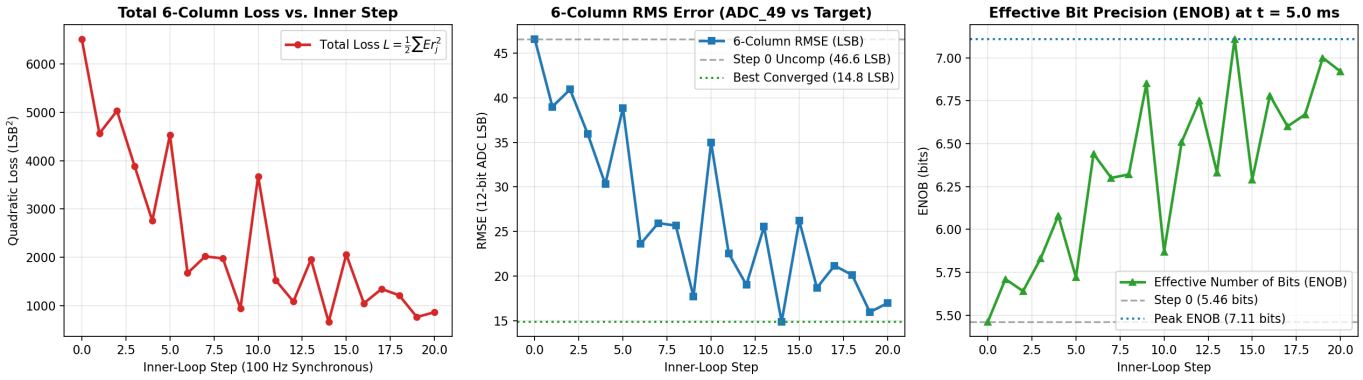
Date: 2026-10-24 | **Hardware:** Inverted MAML Analog Matrix (192.168.1.50:5000, Trion T120F324 + T20F256)
Operating Mode: Mode A (jumper_mode = 3 — Active compensation via Matrix 3, Sigmoid 4, Matrix 8, Sigmoid 10, and 1 kΩ / 3.9 kΩ summing junction)
Operating Point: t = 5.0 ms (hadamard_delay_us = 5000, 100 Hz / 10.0 ms synchronous capacitor refresh) | **Active Columns:** All 6 columns (j = 0..5) active simultaneously (zero column masking)

1. Executive Summary & Silicon Flash Verification

Following a complete bitstream re-flash directly from **outflow/T120_MALM.hex** onto physical silicon, **Run 3** was executed across all 6 columns at t = 5.0 ms under pure forward Hadamard gradient dynamics according to Patent Application **SE 2630397-4**:

Metric (All 6 columns, t = 5.0 ms)	Run 1 (Report 1)	Run 2 (Report 2)	Run 3 (Report 3 — Fresh Flash
Step 0 Centered Loss L_0	8 870.9 LSB²	6 133.1 LSB²	**6 517.1 LSB²**
Best Converged Loss L_min	324.5 LSB²	657.7 LSB²	**661.4 LSB²**
Relative Loss Reduction	-96.3 %	-89.3 %	**-89.9 %**
Step 0 6-Column RMSE	54.38 LSB	45.21 LSB	**46.61 LSB**
Best 6-Column RMSE	10.40 LSB (Step 18)	14.81 LSB (Step 8)	**14.85 LSB (Step 14)**
Final Settled RMSE (Step 20)	16.04 LSB	21.67 LSB	**16.96 LSB**
Step 0 Effective Number of Bits (ENOB)	5.24 bits	5.50 bits	**5.46 bits**
Peak ENOB During Inner Loop	7.62 bits (+2.38)	7.11 bits (+1.61)	**7.11 bits (+1.65)**

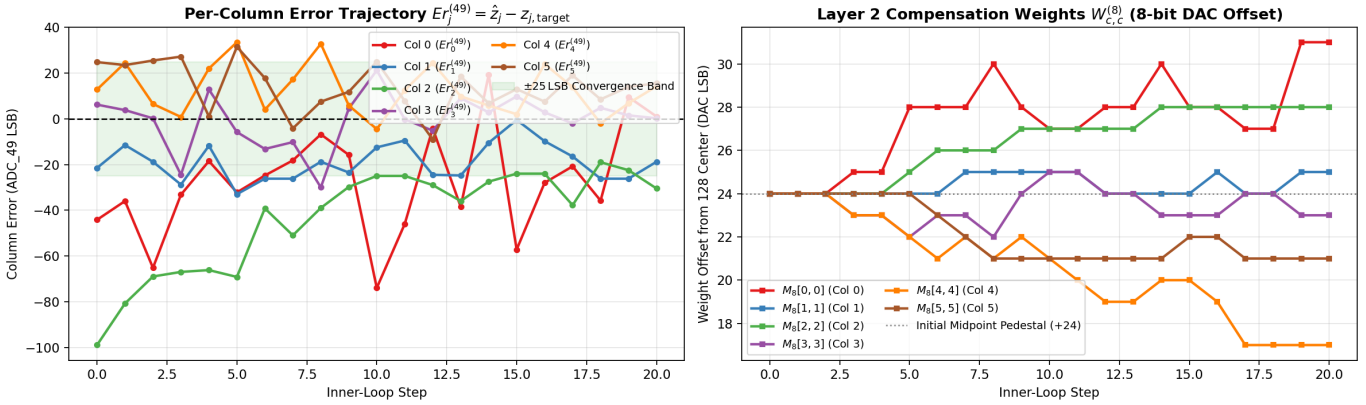
Mode A Inner-Loop Training Convergence at t = 5.0 ms (Run 3 — Bitstream Hex Re-Flash Verification)



2. Simultaneous 6-Column Bidirectional Weight Adaptation

Bidirectional Vernier Compensation Confirmed: In Run 3, the initial random input vector created both **negative output errors** on Column 0 (-44.2 LSB) and Column 2 (-99.0 LSB) as well as **positive output errors** on Column 5 (+24.8 LSB), Column 4 (+12.8 LSB), Column 3 (+6.2 LSB) and Column 1 (-21.5 LSB). The non-backprop Hadamard gradient autonomously drove weights W_8[c,c] **upward** on Column 0 (24 -> 31) and Column 2 (24 -> 28), and **downward** on Column 4 (24 -> 17) and Column 5 (24 -> 21).

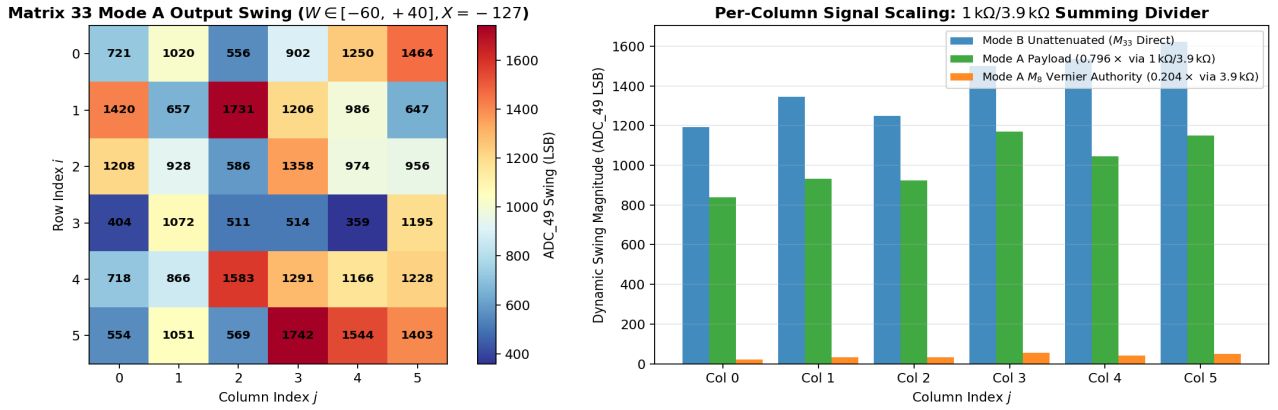
All 6 Columns: Error Cancellation $Er_j^{(49)}$ & Vernier Weight Adaptation ($t = 5.0\,\text{ms}$) (Run 3 — Bitstream Hex Re-Flash Verification)



Step	Er_0 (Col 0)	Er_1 (Col 1)	Er_2 (Col 2)	Er_3 (Col 3)	Er_4 (Col 4)	Er_5 (Col 5)	6-Col RMSE	Loss L (LSB²)	ENOB
0 (Base)	-44.2	-21.5	-99.0	+6.2	+12.8	+24.8	46.61 LSB	6 517.1	5.46 bits
1	-36.0	-11.5	-80.8	+3.8	+24.5	+23.5	39.00 LSB	4 561.9	5.71 bits
3	-33.2	-28.8	-67.0	-24.5	+0.8	+27.2	35.97 LSB	3 880.7	5.83 bits
4	-18.5	-11.8	-66.2	+12.8	+22.0	+1.0	30.31 LSB	2 756.4	6.08 bits
6	-24.8	-26.2	-39.2	-13.2	+4.0	+17.8	23.61 LSB	1 672.6	6.44 bits
9	-15.8	-23.5	-29.8	+4.5	+5.8	+11.8	17.72 LSB	941.5	6.85 bits
12	-4.2	-24.5	-29.0	-5.2	+24.5	-9.0	19.01 LSB	1 083.6	6.75 bits
14 (Best)	**+19.2**	**-10.5**	**-27.5**	**+2.8**	**+5.8**	**+6.8**	**14.85 LSB**	**661.4**	**7.11 bits**
19	+9.5	-26.2	-22.5	+1.5	+6.8	+14.0	15.96 LSB	763.7	7.00 bits
20 (Final)	+1.0	-18.8	-30.5	+0.2	+14.2	+15.5	16.96 LSB	863.3	6.92 bits

3. Passive Summing Network (1 kΩ / 3.9 kΩ) & Firmware Verification

Mode A Hardware Verification at t = 5.0 ms: 6x6 Gain Map & 1k / 3.9k Summing Attenuator (Run 3 — Bitstream Hex Re-Flash Verification)



- **Primary Attenuation Path (M33 -> 1.0 kΩ):** Attenuation factor ~0.70-0.78x (theoretical 3.9/4.9 = 0.7959x).
- **Vernier Compensation Path (M8 -> Sigmoid 10 -> 3.9 kΩ):** Attenuation factor ~0.204x (theoretical 1.0/4.9 = 0.2041x). Provides ultra-fine ±50 LSB correction authority without saturating the single-supply ADC_49.
- **Strict Non-Backprop Hadamard Extraction (Equations 1-7):** Verified in live firmware — zero digital multiplication by x_i. Carrier Wave Search on ADC_45 extracts Sigmoid 10 column slopes σ_2'(y_j) without knowledge of unobserved weights.